

Saddiq Rupani

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EDUCATION

Georgia Institute of Technology

Bachelor's of Science in Computer Science, Minor in FinTech

Atlanta, GA

Expected Graduation: May 2027

- **Concentrations:** Artificial Intelligence and Information Internetworks
- **GPA:** 3.91/4.00
- **Awards:** Dean's List, Faculty Honors, Zell Miller Scholar, Platinum FCU Scholarship Winner
- **Relevant Coursework:** Data Structures and Algorithms, Object-Oriented Programming, Artificial Intelligence, Databases

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, HTML/CSS, SQL, C

Frameworks & Libraries: React, Node.js, FastAPI, LangGraph, JUnit, Pandas, NumPy, Matplotlib, Firebase

Tools & Platforms: Git, GitHub, Docker, Rest API, Android Studio, Linux/Unix, Tableau, Power BI

Certifications: AWS Cloud Practitioner Essentials, Lean Six Sigma Green Belt

EXPERIENCE

Caterpillar

AI Software Engineering Intern

Chicago, IL

May 2026 – Present

- Engineering **Sidekick**, a **multi-agent AI RAG** knowledge pipeline, in **Python** using **LangGraph** and **LangChain** across enterprise data sources, including finance, operations, & HR, integrating **4,000+** user agents and supporting **6,000+** users.
- Developing **dynamic model selection** for Sidekick, integrating **GPT**, **Claude**, and other **LLMs** via **Azure AI Foundry API**, implementing tool binding, prompting, and backend switching, deploying **8** configurations and reducing costs.
- Building an **agentic** code migration pipeline using **GitHub Copilot** and **Azure DevOps MCP** to scan legacy codebases, generate **100+** user stories, populate **GitHub Spec Kit**, and scaffold **Java/PostgreSQL**, reducing migration time by **80%**.

Delta Air Lines

Data Engineering Co-Op

Atlanta, GA

January 2025 – May 2026

- Developed **Machine Learning** models in **Python** using **XGBoost** and **Random Forest** to predict flight catering needs for **4,000+** daily flights, integrated **Teradata** and **PostgreSQL** data sources and achieved **~70%** classification accuracies.
- Established **AWS** cloud infrastructure, automated **50+** workflows using **Glue**, **Lambda**, and **S3**, reduced manual processing time by over **75%** and integrated **Bedrock** to enable **Agentic AI** capabilities for automated data analysis and insights.
- Engineered **Python-based ETL** data pipelines to process **10M+** rows of large-scale data from **SAP** and **Oracle** databases using **SQL**, automated **Tableau & Power BI** dashboards and increased demand forecast accuracy by **8-12%**.

College of Computing at Georgia Tech

Teaching Assistant

Atlanta, GA

May 2024 – December 2025

- Instructed and mentored **500+** students per semester in **CS 1301: Intro to Computing in Python** across **4** semesters, led recitations, **10** weekly office hours, and personalized one-on-one academic support to reinforce programming knowledge.
- Taught advanced CS concepts including **Data Structures**, **Algorithms**, **Object-Oriented Programming**, and **Time & Space Complexity**, graded over **1,000** exams overall and drove a **6%** improvement in average exam scores course-wide.

PROJECTS

RepIQ | React, FastAPI, Supabase, OpenAI API

June 2026 – Present

- Building a cross-platform fitness app using **React Native**, **Expo**, and **TypeScript** with a **FastAPI + Supabase** backend, featuring workout logging, user authentication, and a gym-specific social feed with group accountability challenges.
- Engineering an **AI form analysis** pipeline using **MediaPipe** pose estimation to score lift technique in real-time and integrating **OpenAI API** to generate adaptive post-set quizzes reinforcing personalized form corrections.

Coding Challenge Generator | Python, React, FastAPI, Clerk

September 2025 – November 2025

- Developed a **full-stack app (FastAPI, React)** integrating **OpenAI API** to generate coding challenges across **3** difficulty tiers, all secured via **Clerk Auth** and **REST API** endpoints backed by a **PostgreSQL** database for challenge storage.
- Deployed the application across **3** managed cloud platforms, including **Vercel**, **Render**, and **Supabase**, while implementing a per-user daily quota system capped at **50** requests, with automated provisioning via **Clerk webhook** event handling.

LEADERSHIP

Ismaili Volunteering

Volunteer

Atlanta, GA

August 2024 – Present

- Directing events as a rep with the Ismaili Student Network (ISN), uniting **60+** Ismaili students at Georgia Tech and organizing collaborations that reached more than **3,000** students nationwide, strengthening cultural engagement and presence.
- Working with **50+ volunteers** to coordinate daily prayers and ceremonies for over **400** attendees, ensuring smooth operations, community engagement, and consistent event execution across gatherings.
- Teaching classes of more than **30 K-12 students** about the **1,400-year history** of Ismaili faith, creating interactive lessons.